



PRODUCT NAME

Industrial-grade Mobile Test Formula

Description

A high-viscosity gel designed for industrial applications, ensuring robust performance with mobile-friendly functionalities in mind. This formulation is engineered for applications requiring a stable pH level and cost-effectiveness.

Technical Specifications

pH Level:	6
Viscosity:	High viscosity
Shelf Life:	24 months when stored properly
Batch Size:	100-500 kg
Processing Time:	2 hours
Temperature:	Room temperature (20-25°C)
Storage Conditions:	Store in a cool, dry place away from direct sunlight
Equipment:	Mixing vessel with agitator, pH meter
Certification:	ISO 9001:2015

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Deionized Water	Aqua	80.0%	Solvent and base for gel formation
2	Carbomer	Carbomer	1.0%	Thickening agent to create high viscosity
3	Triethanolamine	Triethanolamine	0.5%	pH adjuster and neutralizer for Carbomer
4	Glycerin	Glycerin	5.0%	Humectant to maintain moisture
5	Hydroxyethylcellulose	Hydroxyethylcellulose	1.5%	Secondary thickening agent for additional viscosity
6	Preservative	Phenoxyethanol	0.8%	Prevents microbial growth
7	Sodium Hydroxide	Sodium Hydroxide	0.2%	pH adjustment
8	Fragrance	Fragrance	0.5%	Adds scent (optional)
9	Polyacrylate Crosspolymer-6	Polyacrylate Crosspolymer-6	10.0%	Enhanced thickening and stability
10	Caprylyl Glycol	Caprylyl Glycol	0.5%	Preservative booster and skin conditioning agent
Total			100%	

Manufacturing Process

Phase 1: Water Phase

1. Weigh and add deionized water into the main mixing vessel.
2. Disperse hydroxyethylcellulose in the water under gentle agitation until fully hydrated.

Phase 2: Gel Formation Phase

1. Add carbomer to the water phase slowly while mixing to avoid clumping.
2. Once carbomer is fully dispersed, neutralize the solution by adding triethanolamine, adjusting pH to 6.

Phase 3: Addition Phase

1. Incorporate glycerin into the mixture as a humectant.
2. Add the polyacrylate crosspolymer-6 under constant stirring for enhanced viscosity.
3. Introduce preservative, caprylyl glycol, and optional fragrance into the formulation.

Required Equipment

Mixing vessel with agitator, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: ISO 9001:2015

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply the gel as needed, ensuring an even application over the surface. Suitable for industrial applications with high-viscosity requirements.